

Geonhee Kim

Web development · Computer vision · Operations automation

geonhee@forblune.com · github.com/forblune · forblune.github.io

Summary

I write down whether what I built **is up right now**, and **what it is not**, in the same table. I train computer vision models and ship them as far as an app, I build and run responsive sites, and I find **the places where something already running broke quietly** with evidence anyone can reproduce. Every number below is one I measured myself. I do not present self-initiated work as paid client work.

Selected projects

RePET Vision — automatic sorting of clear PET bottles · Sep 2025 · Gangwon AI Idea Competition Two-class object detection, clear and coloured PET. 700 images collected through Roboflow, augmented to **1,726** (train 1,539 / val 126 / test 61), YOLOv8n · 640px · 40 epochs · batch 16. **mAP50 approx. 91%** · **mAP50-95 approx. 52%**. Built through to an Android Studio app with live camera recognition and confidence display. Attached an economic case using Gangneung city recycling rates, 330 → 538 KRW/kg.

K2J — smoking detection on CCTV · Jul 2025 · CKU summer AI project Single-class detection on park CCTV footage. YOLOv5 · PyTorch · 640px · 50 epochs. **mAP@0.5 approx. 0.97** · **mAP@0.5:0.95 approx. 0.68** · **Precision approx. 0.90** · **Recall approx. 0.98**. I owned the progress record, wrote the video boundary-correction code, and built the frame-interval feature.

MindHub — mental health care platform · Jun 2026 to present Patient daily records → AI clinical summary → handoff to the clinician, and CBT homework prescribed back the other way. React · Supabase · Kakao login · Solar (Upstage) LLM. Scoped deliberately as **recording, summarising and handoff, not diagnosis**.

Eight live sites — addresses and boundaries are tabulated at forblune.github.io. Web Care (pricing, refunds, enquiry, checkout), Style Gallery (8 directions, 24 pages, 8 in English, single-file HTML with no framework), Mellow Room · OpsFlow · ClientFlow · ServiceOS (self-initiated demos, fictional data). All of them respond right now.

Problems solved

Cases from sites that are live. **One of the four below is an incident I caused, and I wrote that into the document as it happened.**

Enquiries were not arriving and the page looked fine — a static-only deploy dropped the server function, and the enquiry API returned **405 for about four hours**. Separating **400** (bad request) from **405** (path exists, method gone) settled "since when, and because of what." After recovery I froze deploy verification into a procedure — check the deployment URL, not the domain, because a CDN cache will return 200 for a file that is already deleted.

Nobody knew where the current version was — the source of a live site was in no git repository. Hashing eight local candidates against live gave **0 matches**. I fixed the live output as canonical and made the deploy one-directional.

Someone else's outage became our outage — a free external image service was hotlinked directly, so half the page turned into grey boxes during their incident. **243 external hotlinks → 0**, with 234 files selected as CC0 and moved to self-hosting.

The real cause of "it looks AI-generated" — usually the fundamentals rather than the content. I quantified Korean breaking mid-word using `Intl.Segmenter` and `Range.getClientRects()`, applied it across every live site and 25 example pages, and brought it to **0**.

Awards

Date	Award
Feb 2024	Grand Prize, 2024 CKU pre-university bootcamp, "Building a chatbot with generative AI" · hosted by Catholic Kwandong University · team 4 leader · three-day team build (MIT App Inventor)

Date	Award
Jul 2024	Encouragement Award, 22nd Next-Generation Global Business School · hosted by World-OKTA · 1–6 Jul 2024 · team presentation smart WALK
2025	Encouragement Award, academic conference presentation on space logistics · sole presenter, "Earth–Moon–Mars three-stage sustainable deep space logistics"

Business and operations

Open-market advertising operations · Jan–Mar 2025 — managed ad groups for **around 1,700 products** across Gmarket, Auction and 11st. Improved ROAS through ad on/off decisions, category matching, and keyword and daily-budget adjustment, and made the calls on restructuring the catalogue.

Naver Smart Store, sole operator · Oct 2024 to present — ran the full cycle under my own business registration: sourcing, listing, orders, shipping, exchanges and customer service, and settlement.

Other — NYPC Code Battle entrant (Jul 2025, competitive algorithm strategy) · RowCraft (May 2026 to present, a solo data-cleaning business on Fiverr, with an SOP built on pandas and rapidfuzz plus a Slack bot) · BTCUSDT paper-trading bot running continuously on a Raspberry Pi · water purifier lever designed and 3D printed from scratch · Raspberry Pi 5 and Arduino sensor work.

How I work

I do not fix by guessing. I build **a reproducible boundary** that settles the cause first. Code compiling and code working for a user are different things, so I confirm the fix on the real screen. I leave why it happened and how to check it in the repository, so the same incident does not happen twice. **I write down my own mistakes too** — hiding them costs the credibility of everything else I report.

Education and skills

Education	Catholic Kwandong University, Air Transport and Logistics · enrolled Mar 2024 · Baekyang High School (gifted programme, 115 hours of mathematics and science)
Training	ICT Innovation Square Capital Region — Vibe Coding web project course , completed · 1–22 Jun 2026 (80 hours) · National IT Industry Promotion Agency (NIPA) CKU AI Academy advanced specialisation (computer vision, NLP, multimodal) · Jun–Jul 2025
Computer vision	YOLOv5/v8 · Roboflow · OpenCV · PyTorch · Google Colab
Web and app	HTML · CSS · JavaScript · TypeScript · React · Supabase · Android Studio
Deploy and infrastructure	Cloudflare Workers/Pages · D1 · GitHub Actions · Git · Raspberry Pi
Quality	Responsive layout · browser QA · accessibility fundamentals · Korean typography